Start

Initialization: $\mathcal{D}_0$, Inputs: $\mathcal{C}$, T, β, T_{max}

Data Core: $\mathcal{D}_t = \{(x_i, y_i)\}$

1. BCS Module

Sparse Bayesian Learning (Trunc. DCT)
→ Posterior (μ, Σ)

2.BO Module

Fit GP on $\mathcal{D}_t$
→ BO-GP (μ, σ²)

Posterior (μ, Σ)

BCS-GP
(Derived from μ, Σ)

Posterior Σ

BO-GP

BCS-GP

BCS Exploration Score (ES)
(Derived from Σ)

3. Surrogate Fusion

Uncertainty-aware weighting
→Combined GP (Comb-GP)

Comb-GP

Compute Expected Improvement (EI)

ES value

EI value

4. Acquisition Fusion

Normalize and sum EI & ES
Fused Acquisition Function

Maximize Fused Acquisition
→ Select next point x_{t+1}

Observe y_{t+1}, Update $\mathcal{D}_{t+1}$

Iteration Loop

Yes

$t < T_{max}$?

No

Output: best observed (x^*, y^*)

End